

HealthConnect Final Analytics & Decision Support Package

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1. EXECUTIVE SUMMARY

The HealthConnect Clinic project addresses high appointment no-show rates (48.46%) that cause severe operational idle capacity and revenue leakage. Over Weeks 5 through 8, the Data Analytics track audited 5,000 clinic records, executed statistical hypothesis tests, validated non-linear segment thresholds, and delivered engineered feature matrices to downstream Data Science and ML pipelines.

Key operational insights confirm that booking lead times exceeding 21 days and clinic wait times exceeding 30 minutes are the primary drivers of patient drop-off. Implementing mandatory SMS reminder workflows yields a +6.92 percentage point increase in patient attendance over unreminded cohorts.

2. VALIDATED BASELINE KPIS & METRICS MATRIX

KPI Metric / Name	Calculated Value	Benchmark / Standard	Status	Key Operational Finding
Evaluated Records	5,000	N = 5,000 Dataset	PASSED	Complete record audit with zero missing outcome values.
Baseline No-Show Rate	48.46%	2,423 / 5,000 No-Shows	PASSED	Confirmed programmatically via Python assertion checks.
SMS Attendance Peak	49.60%	vs No Reminder (42.68%)	PASSED	SMS outperformed Email (46.53%) and WhatsApp (44.60%).
Lead Time Gap	34.53 Days	vs Attended (24.52 Days)	PASSED	10.01-day mean extension for non-attending patient cohort.
False-Negative Reduction	12.00%	Data Science Pipeline Retest	PASSED	Categorical binned features improved ML classification accuracy.

3. CORE BUSINESS INSIGHTS & OPERATIONAL DRIVERS

- The 21-Day Lead Time Risk Cliff:** Patient attendance remains relatively stable for appointments booked within 0–21 days (~36.4%–37.2% no-show rate). Once lead times cross 21 days, no-show rates jump to over 63%, identifying long lead intervals as the single primary source of scheduling loss.
- Communication Channel Superiority:** SMS reminders consistently outperform alternative channels, providing an immediate +6.92% lift in compliance compared to sending no reminders (42.68%).
- Queue Wait Time Attrition:** Clinic wait times exceeding 30 minutes correlate strongly with patient drop-offs, disproportionately impacting senior patients (aged 56+).

4. ACTIONABLE STRATEGIC RECOMMENDATIONS

- Policy Recommendation 1 (Dynamic Overbooking Caps):** Restrict overbooking buffers exclusively to booking slots scheduled further than 21 days out to offset the 63%+ expected no-show rate.
- Policy Recommendation 2 (Automated SMS Alerts):** Deploy mandatory 48-hour automated SMS reminder sequences for all appointments, prioritizing long lead-time bookings.
- Policy Recommendation 3 (Senior Queue Fast-Tracking):** Establish automated fast-track dispatch protocols for senior patients (aged 56+) whenever clinic queue wait times breach 25 minutes.

5. CROSS-TRACK CONTRIBUTION & INTEGRATION RECORD

- Data Analytics → Data Science:** Exported `HealthConnect_Feature_Engineered_Segments_Validated.csv` containing binned categorical features (`lead_time_bracket` , `age_group` , `high_wait_flag`). These inputs resolved high baseline false-negative errors in the predictive model.
- Data Analytics → Project Management:** Provided validated KPI benchmarks, risk threshold charts, and operational recommendation frameworks to inform project closure metrics and executive presentation alignment.

6. ANALYTICAL LIMITATIONS & FUTURE ENHANCEMENTS

- **Data Limitations:** The historical dataset lacks real-time timestamps for patient check-ins and dynamic rescheduling logs.
- **Future Work:** Future iterations should incorporate real-time clinic capacity feeds and patient socio-economic mobility factors to improve predictive accuracy.